



## Running a small business

### Task A: Profit in percentages

1) Fill in the blanks for this small business' first few years of results and comment on how their profit margin is changing.

|            | Gross profit | Revenue        | Gross profit margin |
|------------|--------------|----------------|---------------------|
| Year one   | £0           | £12 000        | 0%                  |
| Year two   | £1000        | £20 000        | 5%                  |
| Year three | <i>£6000</i> | £48 000        | 12.5%               |
| Year four  | £20 000      | <i>£50 000</i> | 40%                 |
| Year five  | £24 500      | £56 000        | <i>43.75%</i>       |

*You might have reported, "Whilst revenue increases are slowing the gross profit margins have increased to a healthy level".*

2) Calculate the gross profit margin for each of these items for this new paddleboard shop.

$$\text{Gross profit margin (\%)} = \frac{\text{Selling price} - \text{Cost price}}{\text{Selling price}} \times 100$$

a) 10 ft paddleboard: Selling price £575, cost price £373.75      35%

b) 12 ft paddleboard: Selling price £650, cost price £390      40%

c) Carbon paddle: Selling price £110, cost price £82.50      25%

d) Buoyancy aid: Selling price £60, cost price £34.50      42.5%

e) Comment on which product is not achieving their 30% gross profit margin target and advise on what they should do.

*e) The carbon paddle is not achieving 30% G.P. The business needs to either find a cheaper supplier, raise the selling price, or stop selling this product.*



3) a) By completing a spreadsheet or otherwise, calculate the minimum selling price and gross profit margins for this bicycle shop's products.

| Product                      | Cost price | Min. selling price | Actual selling price | Gross profit % |
|------------------------------|------------|--------------------|----------------------|----------------|
| Tyres (min 20% g.p.)         |            |                    |                      |                |
| Road bike tyre               | £14.00     | £17.50             | £19.50               | 28             |
| Mountain bike tyre           | £16.00     | £20.00             | £22.00               | 27             |
| Mountain bike (extra wide)   | £19.00     | £23.75             | £25.00               | 24             |
| Anti-puncture tyre           | £21.00     | £26.25             | £25.00               | 16             |
| Pedals (min 30% g.p.)        |            |                    |                      |                |
| Studded road bike (pair)     | £26.00     | £37.14             | £38.00               | 32             |
| Studded mountain bike (pair) | £28.00     | £40.00             | £40.00               | 30             |
| Cleated road bike (single)   | £18.00     | £25.71             | £26.00               | 31             |
| Cleated road bike (pair)     | £34.50     | £49.29             | £48.00               | 28             |
| Brakes (min 45% g.p.)        |            |                    |                      |                |
| Pads (pair)                  | £4.20      | £7.64              | £13.00               | 68             |
| Cable (single)               | £1.20      | £2.18              | £4.75                | 75             |

b) Comment on any products not achieving minimum gross profit margins.

*Anti-puncture tyres and cleated pedals (pair) do not achieve minimum G.P. margins.*

c) Advise the business as to which products they should place most prominently in their shop.

*They should place brake cables most prominently in their shop.*



## Task B: Tracking performance

1) Here are the first three years of sales for a new small business.

| Month              | Jan | Feb  | Mar | Apr  | May  | Jun  | Jul  | Aug  | Sep  | Oct  | Nov  | Dec  |
|--------------------|-----|------|-----|------|------|------|------|------|------|------|------|------|
| Y1 revenue (000's) | 8.0 | 8.4  | 9.0 | 8.3  | 11.0 | 10.7 | 9.8  | 11.1 | 12.5 | 17.0 | 27.2 | 38.4 |
| Y2 revenue (000's) | 7.6 | 7.8  | 9.1 | 10.2 | 9.7  | 8.8  | 10.3 | 10.1 | 14.2 | 21.0 | 38.4 | 49.4 |
| Y3 revenue (000's) | 9.7 | 10.1 | 9.9 | 12.3 | 9.5  | 9.8  | 11.4 | 12.5 | 13.6 | 20.7 | 47.0 | 61.6 |

a) Calculate the mean and range for each year and comment on any changes year-on-year.

Year one:

Range: £30 400

Mean: £14 283

Year two:

Range: £41 800

Mean: £16 383

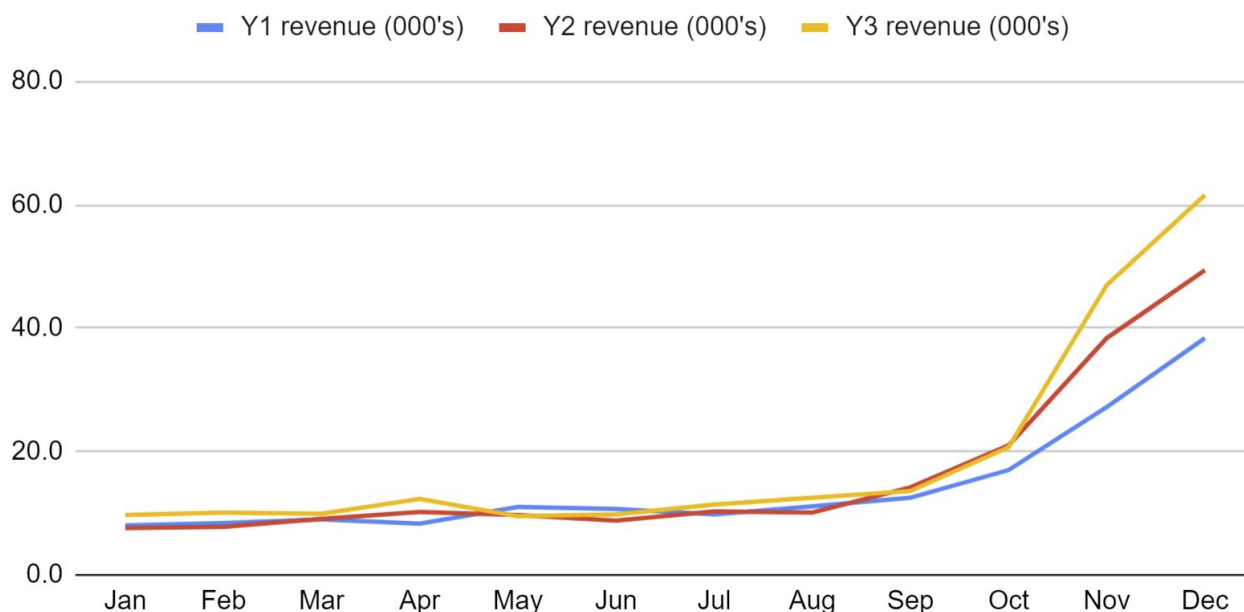
Year three:

Range: £52 100

Mean: £19 008

You might have said, "The mean is increasing so the business is selling more year-on-year. The range is increasing so the sales are becoming less consistent, less predictable month-on-month".

b) Open the [spreadsheet](#), highlight this range of data, and click insert/chart to graph the sales data year-on-year.



c) Comment on any patterns you spot in the data.

You might have said, "The graph shows that there is a seasonal peak to the business' sales in Nov/Dec. Their performance is level year-on-year Jan to Sept but the Oct/Nov/Dec sales are improving with each year".

